Turkish Journal of Agricultural and Natural Sciences 12 (4):979-991, 2025
https://doi.org/10.30910/turkjans.1745369

TÜRK TARIM ve DOĞA BİLİMLERİ DERGİSİ

TURKISH JOURNAL of AGRICULTURAL and NATURAL SCIENCES

www.dergipark.gov.tr/turkjans

Original Article

Demographic Aging in Agriculture: Implications for Employment in Turkish Agriculture

Tuğçe Sarıoğlu [1] ✉

[1]Hatay Mustafa Kemal University, Faculty of Agriculture, Department of Agriculture Economics, Hatay, Türkiye

[1]https://orcid.org/0000-0002-5119-8788

✉: tkiziltug@mku.edu.tr

ABSTRACT

This study aims to analyze the impact of demographic transformation in Türkiye's agricultural sector on agricultural employment. By considering factors such as the increasing proportion of elderly individuals in the agricultural labor force, the withdrawal of younger generations from the sector, and structural changes like mechanization, an econometric analysis was conducted using the ARDL (Autoregressive Distributed Lag) model for the 2000–2024 period. Within the analysis, the aging index, total agricultural area, and number of tractors were used as independent variables, while the agricultural employment rate was modeled as the dependent variable. The findings indicate that the aging index has a statistically significant and negative long-term effect on agricultural employment. While the size of agricultural land positively influences employment, increased mechanization has been observed to reduce labor demand. The results of the error correction model reveal that short-term deviations return to equilibrium within approximately 1.5 years. These results underscore the importance of policies aimed at encouraging young people to engage in agriculture and enhancing the attractiveness of rural areas.

Key words: Agricultural employment, ARDL model, demographic change, rural labor, Türkiye.

Tarımda Demografik Yaşlanma: Türkiye Tarımında İstihdam Üzerine Yansımaları

ÖZ

Bu çalışma, Türkiye'de tarım sektöründe yaşanan demografik dönüşümün tarımsal istihdam üzerindeki etkilerini analiz etmeyi amaçlamaktadır. Tarımsal işgücünde yaşlı nüfusun oranının artması, genç nüfusun sektörden uzaklaşması ve mekanizasyon gibi yapısal değişkenler dikkate alınarak, 2000–2024 dönemi için ARDL (Autoregressive Distributed Lag) modeli kullanılarak ekonometrik bir analiz gerçekleştirilmiştir. Analiz kapsamında, yaşlanma endeksi, toplam tarım alanı ve traktör sayısı bağımsız değişken olarak değerlendirilmiş, tarımsal istihdam oranı ise bağımlı değişken olarak modellenmiştir. Elde edilen bulgular, yaşlanma endeksinin uzun vadede tarımsal istihdam üzerinde istatistiksel olarak anlamlı ve negatif bir etki yarattığını göstermektedir. Tarım alanı büyüklüğü istihdamı pozitif yönde etkilerken, mekanizasyon düzeyindeki artışın emek talebini azalttığı gözlemlenmiştir. Hata düzeltme modeli sonuçları, kısa dönemli sapmaların yaklaşık 1,5 yıl içinde dengeye döndüğünü ortaya koymaktadır. Bu sonuçlar, genç nüfusun tarım sektörüne kazandırılması ve kırsal alanların cazibesinin artırılması yönünde politikaların önemini vurgulamaktadır.

Anahtar kelimeler: Tarımsal istihdam, ARDL modeli, demografik değişim, kırsal işgücü, Türkiye.

INTRODUCTION

Labor is a fundamental component that directly and reciprocally interacts with macroeconomic variables. Therefore, it exhibits a dynamic structure that both influences and is influenced by economic fluctuations. The need for micro-level analyses in addressing macro-level economic problems highlights the importance of sectoral and firm-level investigations to better understand labor dynamics. In this context, conducting in-depth analyses

at the sectoral level is crucial for comprehending the structural characteristics of labor markets (Bayramoğlu and Bozdemir, 2020).

The agricultural sector is considered strategically important not only for food production, but also for rural development, employment generation, contribution to foreign trade, supply of raw materials to industry, its share in national income, and its role in maintaining social balance. Therefore, the structure of the labor force in this sector directly influences the sustainability and productivity of agricultural production, as well as the levels of automation and innovation. In particular, the age distribution of the agricultural workforce plays a critical role in technological adaptation capacity, knowledge transfer, and the continuity of production across future generations. The gradual withdrawal of the younger population from agriculture and the increasing dominance of older individuals may lead to a decline in production capacity, a loss of the sector's attractiveness, reduced profitability, limited integration of technology, and marketing challenges in the long term. The phenomenon of agricultural workforce aging and the gradual withdrawal of young people has been observed not only in Türkiye but also internationally, highlighting a global challenge for sustaining rural production (Farrigan et al., 2024; OECD, 2020; FAO, 2019). For this reason, analyzing the agricultural labor structure based on age groups is crucial for understanding sectoral transformation and developing effective policies. Globally, the strategic role of agriculture extends beyond food production to include rural development, employment generation, and technological innovation (Douwe van der Ploeg, 2010).

Although the concept of labor force is defined in various ways in the literature, according to the International Labour Organization (ILO), the labor force "comprises all persons of working age who provide labor supply for the production of goods and services during a specified reference period" (ILO, 2021; Sarıoğlu, 2022). As of 2024, the labor force aged 15 and over in Türkiye has been recorded as 35.7 million people (TURKSTAT, 2024). Of this population, 32.6 million are employed (TURKSTAT, 2024), which can also be defined as individuals participating in the production process and receiving a wage in return for a specified amount of work (Sarıoğlu, 2022).In terms of sectoral distribution, 14.8% of the employed population (approximately 4.8 million people) work in the agricultural sector, while this figure is 20.7% in industry, 6.6% in construction, and 57.9% in the services se ctor. In 2000, 7.7 million people were employed in agriculture, and among them, individuals aged 15–24 accounted for 22.9% (Sarıoğlu, 2022).

Globally, the share of youth employment in agriculture declined to 30.5% in 2021, while the share of youth employed in services rose to 45.9% (ILO, 2024). These figures clearly indicate that the agricultural sector is losing its appeal among young people worldwide. According to data from the Union of Turkish Agricultural Chambers (TZOB) for 2023, there are 5.16 million registered farmers in Türkiye. Of this population, 82% are men and 18% are women. In terms of demographic distribution, 34% of farmers are aged 65 and over, while 34.6% are between the ages of 50 and 64. In contrast, 26.4% fall within the 33–49 age range, 3.8% are aged 25–32, and only 1% are between 18 and 24 years old. Therefore, the total share of farmers aged 18–32 remains as low as 4.8% (TZOB, 2023). These figures not only reflect the aging of the agricultural workforce, but also highlight the ongoing disengagement of younger generations from agricultural production. This situation underlines the critical importance of retaining young labor in agriculture from the perspective of rural development policies.

Young farmers play a key role in ensuring the multifunctionality of agriculture, achieving social sustainability, and mobilizing the sector's innovative potential (Anonymous, 2013). However, the gradual disengagement of rural youth from agriculture hinders the effective utilization of this potential. In a study by Yılmaz et al. (2020), approximately 75% of the surveyed farmers stated that the average age of those engaged in agriculture is steadily increasing, primarily due to the migration of young people from rural areas to cities. The growing dominance of the elderly population in agriculture has led to a decline in farmers' participation in production activities. About 56% of participants indicated that they did not wish to continue their agricultural practices, while 58% believed that their children or close relatives would not pursue careers in the agricultural sector. Similarly, in a case study conducted in Isparta by Başaranoğlu and Yılmaz (2020), it was found that young farmers were less inclined to engage in farming, which poses a threat to the continuity of agricultural production in rural areas. Akdemir et al. (2021) also noted that with increasing age, farmers tend to invest less in agricultural activities and abandon production types that require higher labor input.

In recent years, the tendency of young people in Türkiye to disengage from the agricultural sector has become increasingly evident. This trend can be attributed to various factors, including the perception of agriculture as a low-income activity, the lack of future security, the declining attractiveness of rural life, and the diversification of employment opportunities in non-agricultural sectors (Yılmaz et al. 2020; OECD, 2020). In a study by Sarıoğlu and Şengül (2024), using Household Labor Force Survey data from 2004 to 2020 and conducting a panel data analysis, it was found that individuals with higher levels of education and income are more likely to leave the agricultural sector. This finding is consistent with the broader literature, which shows that young people living in rural areas often do not want to engage in farming and tend to migrate to urban centers for a variety of

reasons (Kurt, 2006; Olhan, 2011; Aşkın et al., 2013; Cebeci et al., 2014; Yalçın and Kara, 2016; Davran et al., 2017; Sevinç et al., 2018). Studies in Türkiye indicate that higher education and income levels are associated with a higher likelihood of leaving agriculture, reflecting both economic and social determinants of youth disengagement (Kurt, 2006; Olhan, 2011; Sarıoğlu and Şengül, 2024).

In parallel with this trend, the increasing preference of the youth to live in cities contributes to a declining share of young labor in the agricultural sector, which poses serious risks to the sustainability of agricultural production. The withdrawal of young people from agriculture leads to an aging rural population, threatens food security (Yıldız et al., 2017), limits the transfer of knowledge and technology, reduces productivity (Yılmaz et al., 2020), and disrupts the continuity of production. From the perspective of the future of agriculture, this situation can be regarded as a critical vulnerability. The aging of the agricultural labor force has drawn increasing attention from researchers in recent years. Günaydın (2018) reports that the share of the 15–64 age group in the total population is projected to start declining by 2040, while the share of the 55–64 age group is expected to increase from 8.2% to 12.8% by 2050.

The main objective of this study is to examine the changes in the employment structure of the agricultural sector in Türkiye based on age groups and to analyze the determinants of agricultural employment by revealing the impact of the aging process on the sector. In this context, the short- and long-term effects of the aging index, agricultural land size, and mechanization level on agricultural employment were empirically investigated for the period 2000–2024. Using time series data, the ARDL bounds testing approach was applied to assess the cointegration relationships among the variables. By presenting original findings on the effects of demographic change on the agricultural labor force, this study aims to offer significant insights for rural development and agricultural policy-making.

MATERIALS AND METHODS

Data Set

In this study, agricultural employment and labor force data based on age groups for the period 2000–2024 were used to examine aging trends in Türkiye's agricultural sector. The data are derived from employment, agricultural land size and the number of tractors statistics published by TURKSTAT and the International Labour Organization (ILO).

Descriptive statistics for the 25-year period analyzed in this study are presented in Table 1. A natural logarithm transformation was applied to reduce the impact of scale differences among variables and to normalize their distribution. This approach minimizes the magnitude differences between variables. The mean value of the Agricultural Employment Rate (LnTISTH) variable is 3.11, with a standard deviation of 0.26. The range between the minimum and maximum values (2.68–3.62) indicates limited variation. The skewness value is 0.34, suggesting a slightly right-skewed distribution, while the kurtosis value of 2.41 indicates a distribution close to normal but somewhat flat. The Jarque-Bera statistic is 0.85 ($p > 0.05$), which implies that the variable follows a normal distribution.

Agricultural Land (LnTARALAN): The variable has a mean value of 10.57 and a very low standard deviation (0.02), indicating that the series has remained quite stable over time and has not experienced significant fluctuations. The skewness is 0.83, showing a right-skewed distribution, while the kurtosis value of 2.28 suggests a distribution close to normal. According to the Jarque-Bera test ($p = 3.43$), there may be minor deviations from normality, but no substantial distortion is observed.

The Aging Index (LnYSEN) has a mean value of 2.91, with a moderate level of variation across observations (SD = 0.13). The skewness value of 0.61 indicates a distribution that is close to symmetric, while the kurtosis value of 2.12 suggests a shape near normal. The Jarque-Bera statistic is 2.36 ($p > 0.05$), indicating an acceptable level of conformity to the normal distribution.

The variable representing the Number of Tractors (LnTRK) has a mean of 13.98 and a standard deviation of 0.16. The range between the minimum and maximum values (13.75–14.28) shows a narrow distribution. The skewness value of 0.30 suggests a nearly symmetric distribution, while the kurtosis value of 1.87 indicates a slightly flatter shape compared to a normal distribution. The Jarque-Bera statistic is 1.70 ($p > 0.05$), which is not statistically significant, implying that the variable follows a normal distribution.

Table 1: Descriptive statistics

Variables	Unit/ Source	Abbreviations	Mean	Standard Deviation	Min–Max	Skewness	Kurtosis	Jarque-Bera (p)
Agricultural Employment Rate	%/TSI,2025c	LnTISTH	3,11	0,26	2,68-3,62	0,34	2,41	0,85
Agricultural Land	1000 ha/ TSI, 2025a	LnTARALAN	10,57	0,02	10,53-10,62	0,83	2,28	3,43
Aging Index	Million people/ILO, 2024	LnYSEn	2,91	0,13	2,72-3,19	0,61	2,12	2,36
Number of Tractors	Number/ TSI,2025b	LnTrk	13,98	0,16	13,75-14,28	0,30	1,87	1,70

Aging Indicators and Trend Analysis

In order to quantitatively express the aging phenomenon in the agricultural sector, the following indicators were calculated. The age groups used in these indicators were defined based on the study by Farrigan et al. (2024). The Elderly Ratio, Youth-to-Elderly Ratio, and Aging Index used in this study are simple descriptive indicators calculated to illustrate the age structure and aging trend in the agricultural sector. They are not novel indices, but straightforward ratios derived from the employment data for clarity and comparability. First, the elderly ratio was computed to measure the share of individuals aged 55 and over in total agricultural employment. An increase in this ratio over time is considered clear evidence of aging in the agricultural sector (Eq 1).

$$Elderly\ Ratio\ \ (\%) = \frac{Employment_{55-64} + Employment_{65+}}{Total\ Agricultural\ Employment} \times 100 \qquad \text{(Eq1)}$$

The youth-to-elderly ratio was calculated to illustrate the relative dominance of young individuals over elderly individuals in agricultural employment. A decrease in this ratio can be interpreted as a sign that young people are gradually withdrawing from the agricultural sector (Eq 2).

$$Youth - to - Elderly\ Ratio = \frac{Employment_{15-24}}{Employment_{65+}} \qquad \text{(Eq2)}$$

In order to more effectively and comparatively illustrate the aging trend in the agricultural sector, the aging index was calculated in addition to the classical age group ratios. This index represents the percentage ratio of agricultural employment of individuals aged 65 and over to those aged 15–24. (Eq3)

$$Aging\ Index = \frac{Employment_{65+}}{Employment_{15-24}} \times 100 \qquad \text{(Eq3)}$$

Econometric method

The Autoregressive Distributed Lag (ARDL) model was developed by Pesaran and Shin (1999) and Pesaran et al. (2001). A key distinction between this model and traditional cointegration tests such as Johansen or Engle-Granger lies in the fact that the ARDL model does not require all variables in the system to be integrated of the same order. This feature offers significant flexibility, particularly for small sample sizes and for series with different integration orders. Although the ARDL model can theoretically be applied without conducting unit root tests, it is still crucial to verify that none of the variables are integrated of order two [I(2)], as there are no valid critical bounds available for series with second-order integration (Pata et al., 2016). Furthermore, ARDL models are widely preferred in the literature due to their ability to provide unbiased estimates of long-run parameters (Harris and Sollis, 2003; Kavas and Kaya, 2023).

The ARDL model equation is presented as follows (Eq 4).

$$\Delta Y_t = \beta_0 + \sum \beta_{1i} \Delta Y_{t-i}\, m_{\,i=1} + \sum \beta_{2j} \Delta X_{t-j}\, m_{\,j=0} + \beta_3 Y_{t-1} + \beta_4 X_{t-1} + \varepsilon_t \qquad \text{(Eq 4)}$$

In the ARDL framework, the coefficients β_3 and β_4 represent the long-run relationships, while β_1 and β_2 reflect the short-run effects. After determining the appropriate lag lengths to be used in the model, the bounds testing approach is applied to examine whether a long-run relationship exists among the variables. In this context, the null and alternative hypotheses tested are formulated as follows (Eq 5):

$H_0: \beta_1=\beta_2=\beta_3=\beta_4=0$ (Eq 5)

This hypothesis is tested to determine whether the levels and lagged values of the dependent and independent variables are jointly significant. In the ARDL bounds testing framework, the computed F-statistic is compared with the critical values provided by the relevant literature. If the calculated F-statistic exceeds the upper bound of the critical value range, the null hypothesis is rejected, indicating the presence of a long-run cointegration relationship among the variables. Conversely, if the F-statistic falls below the lower bound, the null hypothesis cannot be rejected, suggesting that no long-run relationship exists among the variables in the model (Kavas and Kaya, 2023). Once cointegration is confirmed, the second stage of the ARDL bounds testing approach involves the estimation of the long-run relationship. In this study, the long-run ARDL model is specified as shown in Equation 6.Authors should provide a concise description of experimental site, design, materials and analytical methods and statistical procedures used. This section should be as clear as possible to enable other scientists to repeat the research presented (Eq 6).

$$LnISTH_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i LnTISTH_{t-i} + \sum_{j=0}^{q} \beta_i LnYSEn_{t-j} + \sum_{k=0}^{r} \gamma_k LnTARALAN_{t-k} + \sum_{r=0}^{m} \delta_r LnTRT_{t-r} + \varepsilon_t \quad \text{(Eq 6)}$$

RESULTS AND DISCUSSION

Results

The distribution of labor force and employment in Türkiye's agricultural sector by age group is presented in Figure 1. The first graph evaluates agricultural labor force data, showing that the number of young individuals (aged 15–24) employed in agriculture exhibited a declining trend between 2000 and 2023, with the only increase observed in 2024, where the share rose by 1.72%, reaching 5.49 million individuals. The most significant decline occurred during the 2006–2007 period. This drop can be attributed to a shift of the agricultural labor force into non-agricultural sectors following the 2001 economic crisis, particularly between 2001 and 2007. The 25–54 age group showed an increasing trend when taking the year 2000 as the base, with an overall growth of 65.17% by 2024. The smallest annual increase for this age group occurred in 2004. Although the 2001 crisis also impacted this group, an earlier return to the agricultural sector is observed. This may be because migration decisions in this age group tend to affect not only the individual but also their families, often producing a chain effect (Bronars and Borjas, 1990; Castaneda and Triandafyllidou, 2022). Because migration may lead to family fragmentation, weakening of kinship ties, reduced participation in social relations, and disrupted interaction among family members (IOM, 2012, as cited in Yılmaz and Günay, 2022).The 55–64 age group has followed a steadily increasing trend since 2000. Despite a brief decline between 2003 and 2005, the group grew by 138.56% by 2024 compared to the base year. For the 65+ age group, the data show fluctuations from 2002 to 2011, followed by a more consistent increase beginning in 2012. Compared to the base year of 2000, this age group recorded a 53.33% increase by 2024, reaching 1.11 million individuals. In 2020, a noticeable decline across all age groups occurred due to the COVID-19 pandemic, creating a visible break in the trend on the graph.section can be divided into subsections. The noticeable decline in employment across all age groups in 2020 is consistent with the impact of the COVID-19 pandemic on the Turkish labor market. Aldan et al. (2021) highlights that during the pandemic, labor force participation rates decreased, and employment was particularly adversely affected among vulnerable groups.

The second graph in Figure 1 presents agricultural employment data. Throughout the period, the young population (aged 15–24) exhibits a downward trend. The most significant decline occurred in 2009, followed by a continued decline at a slower rate. By 2024, agricultural employment in this age group had decreased by 2.15% compared to the base year, reaching 4.59 million individuals. The 25–54 age group shows an overall upward trend, with a single decline observed in 2004. While the number of employed individuals in this group was 14.68 million in 2000, it increased by 59.94% to reach 23.48 million in 2024. These findings are consistent with the results of the Ninth Development Plan prepared by the State Planning Organization (DPT), which identified the 20–24 age group as the largest group migrating from rural to urban areas, followed by the 15–19 and 25–29 age groups. The report also noted that the tendency to migrate decreases as age increases (Çakır, 2014). According to employment data, the 55–64 age group experienced the highest growth relative to the base year of 2000, with a 130.62% increase by 2024, reaching 3.36 million individuals. While employment among those aged 65 and over

shows a fluctuating trend, notable increases have been observed after 2017. Compared to the year 2000, this group experienced a 49.86% increase by 2024, also reaching 3.36 million individuals. Similar to the labor force series, the year 2020 reflects a visible break in the data due to the COVID-19 pandemic.

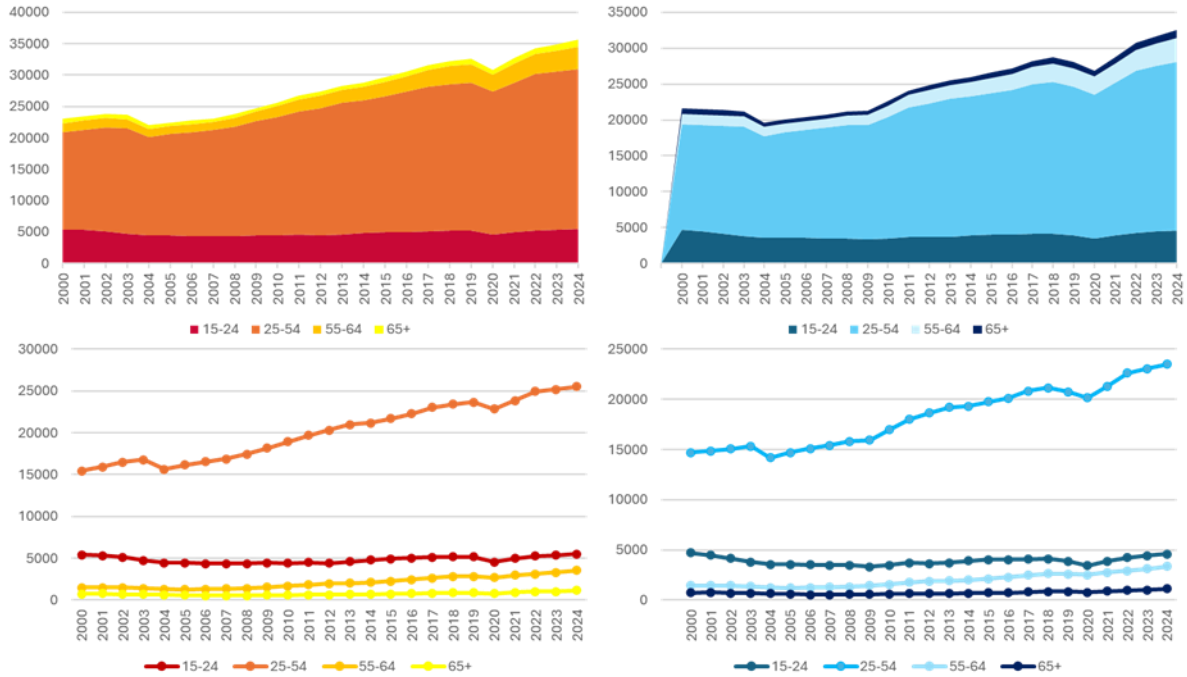

Figure 1. Relative shares of labor force and employment by age groups in the agricultural sector (2000–2024)

Figure 2 presents the 25-year changes in the elderly population ratio (%) and the youth-to-elderly ratio in agricultural employment. In 2000, the proportion of elderly individuals (aged 55 and over) employed in agriculture was 10.21%, which increased to 13.75% by 2024. This indicates that, as of 2024, 13.75% of the agricultural workforce consists of individuals aged 55 and above. While the elderly employment ratio experienced some declines between 2003 and 2010, an upward trend resumed in 2011 and continued thereafter. The youth-to-elderly ratio, on the other hand, decreased from 6.29 in 2000 to 4.11 in 2024, revealing a steady decline in the presence of young individuals in the agricultural sector. This trend suggests that younger generations are increasingly withdrawing from agriculture, while older individuals have become relatively more dominant. Overall, the findings indicate that the aging process in Turkish agriculture is ongoing and reflect the demographic transformation of family farms within the sector.

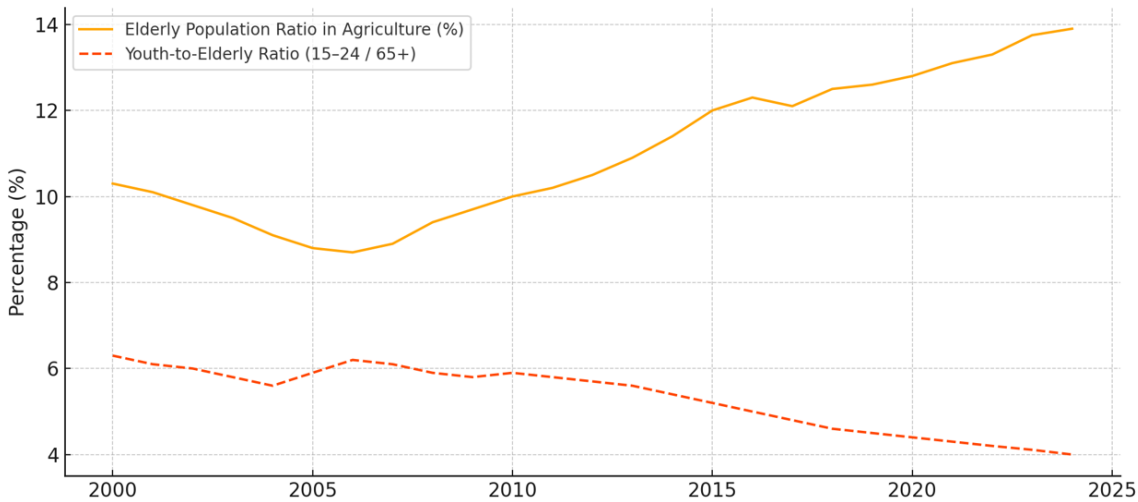

Figure 2. Trends in agricultural aging indicators between 2000 and 2024

To better analyze the aging process in agriculture, the aging index—calculated as the ratio of agricultural employment for individuals aged 65 and over to those aged 15–24—has been utilized. This indicator clearly reveals a declining presence of young individuals and an increasing dominance of the elderly population in the agricultural sector. While the index stood at approximately 15.89% in 2000, it has gradually increased over time and surpassed 20% in recent years. In 2024, it reached its highest level at 24.33%.
This trend indicates that the agricultural sector in Türkiye is undergoing rapid demographic aging, accompanied by a substantial withdrawal of young labor. Compared to traditional age group shares, the aging index serves as a more robust indicator, providing valuable analytical insight into the structural transformation of the agricultural labor force (Figure 3).

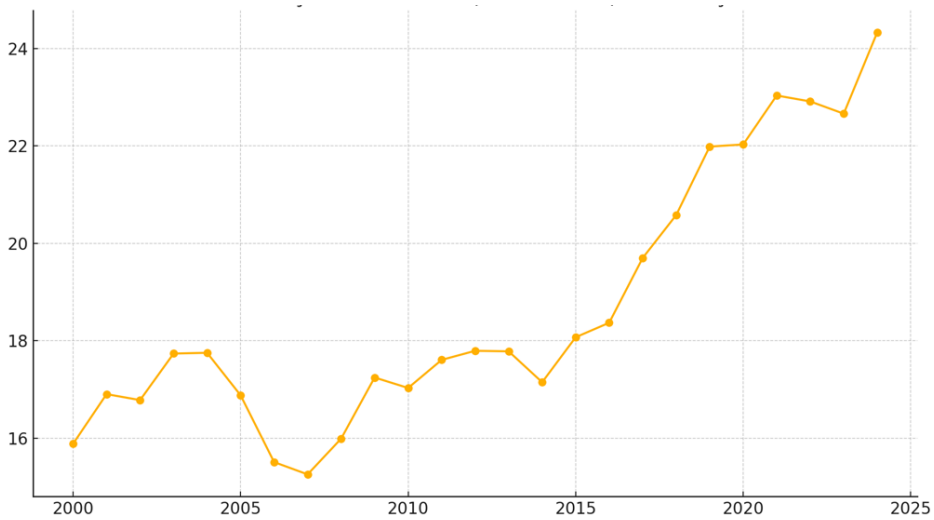

Figure 3. Change in the aging index in agriculture, 2000–2024

The agricultural employment rate, aginig index, agricultural area and number of tractors are illustrated in Figure 4 through level form unit root test graphs. While the agricultural employment rate and agricultural GDP ratio display a decreasing trend, the aging index shows an increasing trend. These patterns suggest that the variables are not stationary at level. To support this preliminary observation, Table 2 presents the results of the Augmented Dickey-Fuller (ADF) Unit Root Test. The ADF test examines the null hypothesis (H_o): "There is a unit root, i.e., the series is non-stationary." If the p-value is greater than 0.05 at level form, H_o is not rejected, and the series is considered non-stationary. When the aging index (LnYSEN) is differenced once, the null hypothesis is rejected ($p < 0.05$), indicating that the series becomes stationary at the first difference. Similarly, for the agricultural employment rate (LnTISTH), the null hypothesis is accepted at level (except in the model with constant), indicating non-stationarity. The agricultural land variable (LnTARALAN) becomes stationary at the first difference ($p < 0.05$), and the same applies to the number of tractors (LnTRK). Since all variables are integrated of order zero [I(0)] or order one [I(1)], and none are integrated of order two [I(2)], the ARDL bounds testing approach is appropriate and valid for examining cointegration relationships.Therefore, the ARDL model is deemed suitable for identifying the existence of a long-run relationship among the variables.

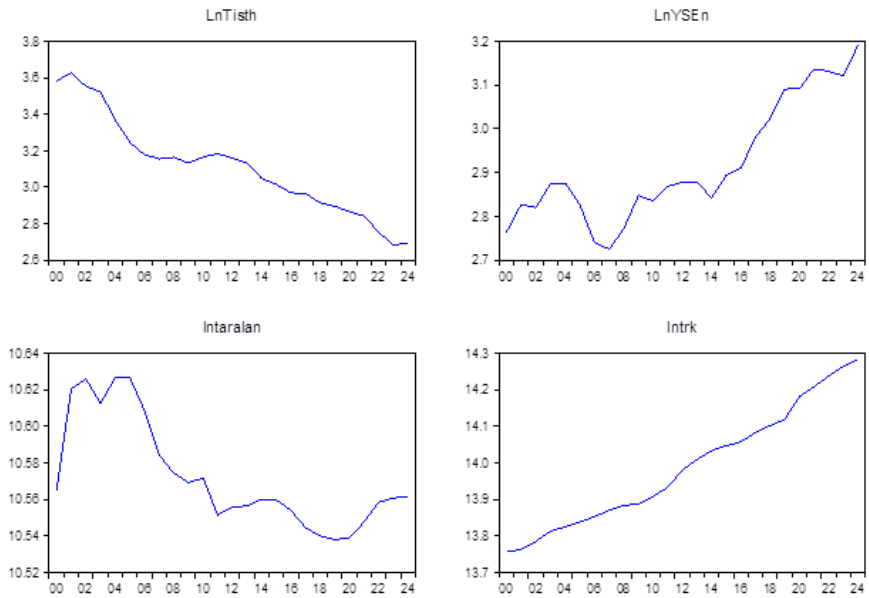

Figure 4. Unit Root Test graphs of variables at level

Table 2. ADF Unit Root Test results

Variables	I(0) (Level Form)					
	None		Constant		Constant and Trend	
	t-statistic	p-value	t-statistic	p-value	t-statistic	p-value
LnYSEn	2.061339	0.9880	0.230972	0.9689	-1.277026	0.8693
LnTISTH	-3.971873	0.0003	-0.846515	0.7872	-4.139699	0.0185
LnTrk	8.638033	1.0000	1.697072	0.9993	-1.408623	0.8319
LnTARALAN	-0.051184	0.6556	-1.205602	0.6547	-2.870497	0.1884
Variables	**I(1)**					
	None		**Constant**		**Constant and Trend**	
	t-statistic	**p-value**	**t-statistic**	**p-value**	**t-statistic**	**p-value**
LnYSEn	-3.503889	0.0012	-3.842641	0.0082	-4.105783	0.0191
LnTISTH	-2.439391	0.0172	-3.716390	0.0108	-3.757103	0.0384
LnTrk	-1.419456	0.1411	-3.931833	0.0067	-4.154347	0.0173
LnTARALAN	-6.239928	0.0000	-6.337203	0.0000	-2.388179	0.3742

In the study, the maximum lag length was set to 3, and the optimal lag structure was determined accordingly. Based on the Schwarz Criterion (SC), the most suitable model for the ARDL test was identified as ARDL(1,3,3,2). According to this model specification, the dependent variable enters the model with one lag, the aging index variable with three lags, the agricultural land variable with three lags, and the number of tractors with two lags. The estimated ARDL model is presented as follows (Eq 7).

$$\Delta\ LNTISTH = \sum \boldsymbol{\beta}_{1i}\Delta LNTISTH_{(t-1)} + \boldsymbol{\beta}_2\Delta\ LNYSEN + \sum\boldsymbol{\beta}_3\Delta\ LNYSEN_{(t-1)} + \sum\boldsymbol{\beta}_4\Delta\ LNYSEN_{(t-2)} + \sum\boldsymbol{\beta}_5\Delta LNYSEN_{(t-3)} + \boldsymbol{\beta}_6\Delta LNTARALAN + \sum\boldsymbol{\beta}_7\Delta LNTARALAN_{(t-1)} + \sum\boldsymbol{\beta}_8\Delta LNTARALAN_{(t-2)} + \sum\boldsymbol{\beta}_9\Delta LNTARALAN_{(t-3)} + \boldsymbol{\beta}_{10}\Delta LnTRK + \sum\boldsymbol{\beta}_{11}\Delta LnTRK_{(t-1)} + + \sum\boldsymbol{\beta}_{11}\Delta LnTRK_{(t-2)} + \boldsymbol{\varepsilon_t} \qquad (Eq7)$$

The diagnostic test results confirm that the estimated ARDL(1,3,3,2) model is statistically valid. The Breusch-Godfrey Serial Correlation LM Test (p = 0.0614) indicates that there is no evidence of autocorrelation in the model. The Breusch-Pagan-Godfrey heteroskedasticity test (p = 0.3631) suggests that there is no problem of heteroskedasticity. The Jarque-Bera normality test (p = 0.8211) confirms that the residuals are normally distributed. In addition, the Ramsey RESET test (p = 0.6802) supports the correct specification of the model and indicates that there is no major misspecification error. The model exhibits a high explanatory power, with R^2 = 0.9955 and adjusted R^2 = 0.9906. The Durbin-Watson statistic was calculated as 2.657, which suggests that the

residuals are free from positive autocorrelation. Although a DW value greater than 2 may raise concerns about negative autocorrelation, the non-significant result of the Breusch-Godfrey test (p = 0.0614) confirms that autocorrelation is not present in this model. Therefore, the overall validity and robustness of the ARDL(1,3,3,2) model are strongly supported.

Table 3. Model diagnostic test results

Diagnostic Tests	Test Statistics	p-value
ARDL(1,3,3,2)		
R^2	0.995518	
Adj R^2	0.990588	
Breusch-Godfrey Serial Correlation LM Test:	4.561790	0.0614
Breusch-Pagan-Godfrey Test	1.277168	0.3631
Jargue-Bera	0.394048	0.8211
Ramsey RESET test	0.425854	0.6802

To evaluate the stability of the ARDL model, CUSUM and CUSUMQ test graphs are presented in Figure 4. Both graphs remain within the critical bounds at the 5% significance level, indicating that the model is generally stable and that the estimated coefficients maintain consistency over time. The results of the CUSUM and CUSUMQ tests suggest that there is no structural break in the series, and the long-run coefficients of the model can be considered reliable.

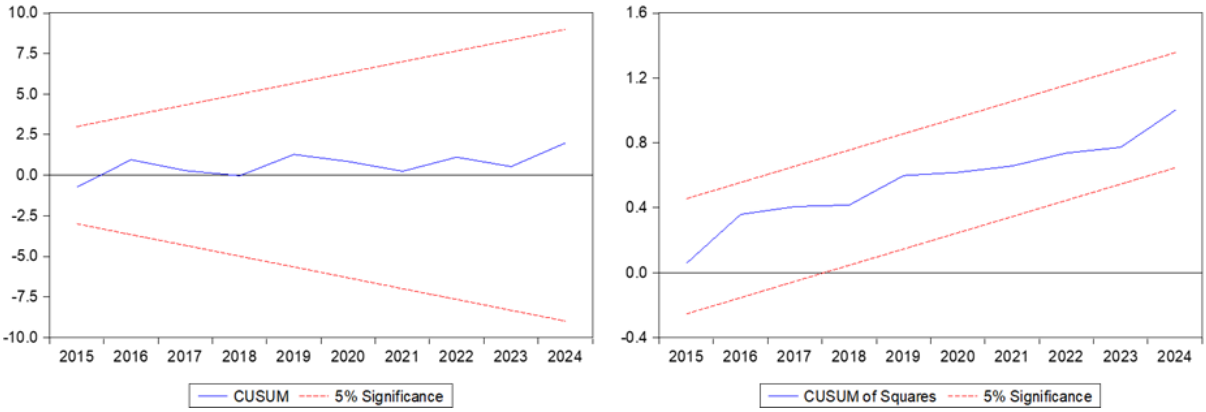

Figure 4. Stability analysis of the model based on CUSUM and CUSUMQ graphs

To determine whether the dependent variable is involved in long-run relationships, the Bounds Test was conducted. The computed F-statistic value was 14.53342, which exceeds the upper critical bound (I(1)) at all conventional significance levels. This result confirms the existence of a long-run cointegration relationship among the variables, indicating a statistically significant association between the dependent and independent variables (Table 4). Accordingly, the estimation of long-run coefficients within the ARDL framework is valid, and the Error Correction Model (ECM) can be applied to analyze short-run dynamics. The ECM results are presented in Appendix 1.

Table 4. ARDL Bounds Cointegration Test results

Number of Regressors (k)	F-statistic	Significance Level	Critical Values Lower Bound I(0)	Upper Bound I(1)
3	14.53342	1%	3.42	4.84
		5%	2.45	3.63
		10%	2.01	3.10

According to the short-run cointegration results derived from the ARDL(1,3,3,2) model, both the aging index (YSEN) and the level of mechanization (TRK) have significant short-term effects on agricultural employment. In the short run, a 1% increase in the elderly population working in agriculture leads to a 0.47% increase in agricultural employment. Similarly, a 1% increase in the level of mechanization results in a 1.67%

increase in employment. The error correction term (ECT) is negative and statistically significant, indicating a high speed of adjustment back to long-run equilibrium. With a correction coefficient of approximately –0.67, about 67% of any short-term disequilibrium is corrected within one period. This implies that deviations from the long-run path return to equilibrium in approximately 1.5 years [1/(ECM) 1/0.67≈1.47years]. These results suggest that short-run deviations are temporary and the system structurally tends to revert to its long-run equilibrium.

Table 5. ARDL Short-Run Cointegration results

Variable	Coefficient	Standard Error	t-Statistic	p-value
D(LNYSEN)	0.472332	0.154411	3.058923	0.0121*
D(LNYSEN(-1))	0.092676	0.310456	0.298515	0.7714
D(LNYSEN(-2))	0.406469	0.183194	2.218788	0.0508
D(LNTARALAN)	0.880082	0.953681	0.922826	0.3778
D(LNTARALAN(-1))	-0.815331	0.962642	-0.846973	0.4168
D(LNTARALAN(-2))	1.218049	0.462403	2.634173	0.0250
D(LNTRK)	1.670654	0.495144	3.374077	0.0071**
D(LNTRK(-1))	1.620071	0.650908	2.488939	0.0320
CointEq(-1)	-0.678232	0.137813	-4.921388	0.0006**

* p<0.05, ** p<0.01.

The long-run coefficients obtained from the ARDL model indicate that the aging index, agricultural land size, and number of tractors have statistically significant effects on agricultural employment. The aging index (LNYSEN) has a statistically significant and negative effect on agricultural employment in the long run ($\beta = -0.786$; $p = 0.0015$). This finding suggests that a 1% increase in the elderly population ratio leads to a 0.78% decrease in agricultural employment over the long term. The agricultural land variable (LNTARALAN) shows a positive and significant effect ($\beta = 1.580$; $p < 0.001$), indicating that a 1% expansion in agricultural land results in a 1.58% increase in long-run employment. Conversely, the number of tractors (LNTRK) has a statistically significant negative effect on employment ($\beta = -0.823$; $p < 0.001$), suggesting that a 1% increase in mechanization leads to a 0.82% decrease in agricultural employment (Table 6).

Table 6. ARDL Long-Run Cointegration results

Variable	Coefficient	Standard Error	t-Statistic	p-value
LNYSEN	-0.785791	0.182028	-4.316871	0.0015*
LNTARALAN	1.579683	0.131684	11.996055	0.0000*
LNTRK	-0.822596	0.131660	-6.247894	0.0001*

* p<0.01

Discussion

Overall, the model reveals that in the long run, factors such as demographic structure, land use, and mechanization significantly shape the level of agricultural employment. According to the long-run results, increases in the aging index and the number of tractors reduce agricultural employment, while an increase in agricultural land contributes positively to employment. In the short run, agricultural employment is influenced by both the elderly population and mechanization, with the system reaching equilibrium in approximately 1.5 years.

These findings are consistent with previous research indicating aging in the agricultural workforce, such as Yıldız et al. (2017), Yılmaz et al. (2020), and Başaranoğlu and Yılmaz (2020). Similarly, Lindsjö et al. (2020), in their study on Malawi, reported that agricultural land was predominantly owned by the elderly, and the lack of sufficient labor reduced productivity. Parallel to these findings, the current study also suggests that agricultural land in Türkiye tends to be concentrated in the hands of the older population. Keskin (2024), in a cointegration analysis, found a negative relationship between mechanization and agricultural employment, reporting that a 1% increase in mechanization led to a 3.74% and 0.42% decrease in employment in the long and short run, respectively. While the long-run negative relationship between mechanization and employment is consistent with our findings, the short-run results differ. Yücel and Çalışkan (2020), using the ARDL bounds test, also identified a negative relationship between mechanization and agricultural employment in their study on productivity and mechanization. Similarly, Shani and Musa (2021) found a negative relationship between agricultural employment and mechanization in Nigeria, aligning with the results of this study.Interpretation of data, related to previous research, should be presented. Results and discussion sections may be combined.

CONCLUSION

This study evaluates the dynamics of agricultural employment in Türkiye within the context of demographic change and the impact of aging on sectoral labor. The findings of the ARDL model indicate that the aging index reduces agricultural employment by 0.78% in the long run. While the expansion of agricultural land contributes to employment growth by 1.58%, increased mechanization reduces labor demand by 0.82%. The error correction term confirms that short-run deviations adjust back to equilibrium within approximately 1.5 years, validating the model's long-run stability.

The results highlight the critical need to attract the country's young population to the agricultural sector to ensure its sustainable development. In this regard, policies that increase the participation of young farmers in production and make rural life more attractive must be prioritized. In addition to broad policy directions, specific measures such as targeted grants of 1–5 million TL for young farmers, low-interest loans with simplified collateral requirements, provision of modern agricultural equipment, digital farm management training, rural infrastructure improvements, and support for youth-led cooperatives can be implemented to encourage youth retention in agriculture. A multidimensional policy approach is essential to facilitate the integration of youth into agriculture and enhance the appeal of rural living. Firstly, financial incentives such as grants, low-interest loans, and land allocation should be provided to young farmers, and specific support mechanisms should be developed for first-time rural entrepreneurs. In addition, knowledge-based agricultural production should be promoted through agricultural training and digital literacy programs. Investments in rural development infrastructure—including internet access, healthcare, education, and social amenities—should be expanded to reverse rural-to-urban migration.

Finally, institutional support should be strengthened in areas such as youth participation in decision-making, cooperative development, innovative agribusiness initiatives, and access to agricultural technologies. These efforts can help mitigate the aging of the agricultural workforce and promote long-term sustainability in the sector.

Declaration of interests

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Author Contributions

Tuğçe SARIOĞLU : All analyses and manuscript writing were performed by TS.

ORCID

Tuğçe SARIOĞLU **http://orcid.org/0000-0002-5119-8788**

Article History	
Submission received:	18.07.2025
Revised:	09.09.2025
Accepted:	12.09.2025

REFERENCES

Akdemir, Ş., Kougnigan, E., Keskin, F., AKÇAÖZ, H., Boz, I., Kutlar, I., Miassi, Y.E., Kusek, G., & Turker, M. (2021). Ageing population and agricultural sustainability issues: Case of Turkey. *New Medit, 20*(4).

Aldan, A., Çıraklı, M. E., Torun, H. (2021). Covid 19 and the Turkish labor market: Heterogeneous effects across demographic groups. *Central Bank Review, 21(*4), 155-163. https://doi.org/10.1016/j.cbrev.2021.12.003

Anonymous, 2013. Policies enabling innovaton, learnng and transition towards sustainability of agriculture. Final Conference of the FarmPath and Solinsa 7th Framework projects. Conference documents, 3 December 2013, Brussels. http://www.solnsa.org/fileadmn/Fles/delverables/Delverable_6.6.pdf

Aşkın, E. Ö. Yayar, R. ve Oktay, Z. (2013). Kırsal Göçün Ekonometrik Analizi: Yeşilyurt İlçesi Örneği. *Cumhuriyet Üniversitesi İktisadi İdari Bilimler Dergisi, 14*(2), 231-252.

Bayramoğlu, Z., & Bozdemir, M. (2020). Definition of Labour Terminology in the Agricultural Sector. *Turkish Journal of Agriculture - Food Science and Technology, 8*(3), 773–783. https://doi.org/10.24925/turjaf.v8i3.773-783.3357

Başaranoğlu, C., & Yılmaz, H. (2020). Genç Çiftçilerin Çiftçilik Yapma Eğilimlerinin ve Çiftçilik Özelliklerinin Belirlenmesi: Isparta İli Örneği. *Tarım Ekonomisi Dergisi, 26*(1), 19-28. https://doi.org/10.24181/tarekoder.588735

Bronars, S. G., Borjas, G. J. (1990). Immigration and the Family. National Bureau of Economic Research. Working Paper. https://www.nber.org/system/files/working_papers/w3509/w3509.pdf?utm_source

Castaneda, R., Triandafyllidou, A. (2022). Migration, Decision Making and Young Families: A Literature Review. https://www.torontomu.ca/content/dam/centre-for-immigration-and-settlement/tmcis/publications/workingpapers/2022_12_Castaneda_R_Triandafyllidou_A_Migration_Decision_Making_and_Young_Families_A_Literature_Review.pdf?utm_source

Cebeci, B. E., Oruç E. ve Kurtaslan, T. (2014). Kırsal Alanda Yaşayan Çocukların Gelecek Yaşamları ve Tarım Sektörü Konusundaki Düşünceler. Türkiye 11. Tarım Ekonomisi Kongresi, (3-5 Eylül, Samsun), 290-299 ss.

Çakır, B. (2014). Tarım Sektörü ve Kırsal Kesimde Yaşanan Çözülmeler Sonucu İşgücü Piyasasında Yaşanan Değişim ve Oluşan İşsizlikte Kamu İstihdam Kurumlarının Rolü. T.C. Çalışma ve Sosyal Güvenlik Bakanlığı Türkiye İş Kurumu Genel Müdürlüğü, Uzmanlık Tezi, Ankara.

Davran, M. K., Özalp, B., Tok, N., Öztornacı, B. (2017). Türkiye'de Kırsal Gençlik Açısından İstihdam ve Tarımsal İstihdamın Geleceği. *Gençlik Araştırmaları Dergisi, 5*(13), 169-199.

Douwe van der Ploeg, J. (2010). The peasantries of the twenty-first century: the commoditisation debate revisited. *The Journal of Peasant Studies, 37*(1), 1-30.

FAO. (2019). Promoting youth engagement and employment in agriculture and food systems. FAO. http://www.fao.org/3/cb5463en/cb5463en.pdf

Farrigan, T., Genetin, B., Sanders, A., Pender, J., Thomas, K. L., Winkler, R., Cromartie, J. (2024). Rural America at a glance (Report No. EIB-282). U.S. Department of Agriculture, Economic Research Service. https://doi.org/10.32747/2024.8722498.ers

Günaydın, D. (2018). Türkiye'nin Demografik Dönüşümü Çerçevesinde İşgücünün Yaşlanma Sorunu. *Yaşlı Sorunları Araştırma Dergisi, 11*(1), 1-13.

Harris, R., ve Sollis, R. (2003). Applied time series modelling and forecasting. Wiley.

ILO, 2021. https://ilostat.ilo.org/resources/concepts-and-definitions/glossary/#L (Accessed date: 15.04.2021)

ILO,2024.https://rshiny.ilo.org/dataexplorer94/?lang=en&segment=indicator&id=EMP_TEMP_SEX_ECO_NB_A&ref_area=TUR (Accessed date: 17.06.2025)

International Organization for Migration (IOM). (2012). Glossary on Migration. 2nd edition, IML Nr. 25, http;//publications.iom.int/bookstore/free/IML25.pdf, in; ACP Observatory on Migration Research Guide for research commissioned by the ACP Observatory on Migration. http;//www.acpmigration–obs.org/ (Accessed date: 25.05.2019)

Kavaz, İ., & Kaya, B. (2023). Türkiye'de Yenilenebilir Enerji Tüketimi, Ekonomik Büyüme ve Co2 Emisyonu Arasındaki İlişkinin Analizi: Ardl Sınır Testi Yaklaşımı. *İşletme Ekonomi ve Yönetim Araştırmaları Dergisi, 6*(2), 219-243. https://doi.org/10.33416/baybem.1313504

Kurt, H. (2006). Göç Eğilimleri ve Olası Etkiler. *Yönetim Bilimler Dergisi, 4*(1), 148-178.

Lindsjö, K., Mulwafu, W., Andersson Djurfeldt, A., & Joshua, M. K. (2020). Generational dynamics of agricultural intensification in Malawi: challenges for the youth and elderly smallholder farmers. *International Journal of Agricultural Sustainability, 19*(5–6), 423–436. https://doi.org/10.1080/14735903.2020.1721237

OECD. (2020). OECD Employment Outlook 2020. OECD Publishing. https://www.oecd.org/content/dam/oecd/en/publications/reports/2020/07/oecd-employment-outlook-2020_19b4fc0d/1686c758-en.pdf

Olhan, E. 2011. Türkiye'de Kırsal İstihdamın Yapısı. Birleşmiş Milletler Ortak Programı "Herkes İçin İnsana Yakışır İş: Ulusal Gençlik İstihdam Programı Antalya Pilot Bölge Uygulaması". http://www.undp.org/content/dam/turkey/docs/projectdocuments/PovRed/ MDG_F_1928/UNDP-TR-YEM_Krsal%20Isthdam.pdf. Erişim: Mart,2020.

Pata, U. K., Yurtkuran, S., & Kalça, A. (2016). Türkiye'de Enerji Tüketimi Ve Ekonomik Büyüme: Ardl Sınır Testi Yaklaşımı. *Marmara Üniversitesi İktisadi Ve İdari Bilimler Dergisi, 38*(2), 255-271. https://doi.org/10.14780/muiibd.281411

Pesaran, M.H. & Shin, Y. (1999). An autoregressive distributed lag modelling approach to co-integration analysis. S. Strom (Ed.) Econometrics and Eonomic Theory in the 20th Century: the Ragnar Frisch Centennial Symposium. Cambridge: Cambridge University Press.

Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationship. *Journal of Applied Econometrics, 16*(3), 289-326

Sarıoğlu, T. (2022). Türkiye'de tarım sektöründe istihdam: Hatay alt bölgesi (TR63) örneğinde tarımda Suriyeli sığınmacı istihdamı. Çukurova Üniversitesi Fen Bilimleri Enstitüsü, Doktora Tezi, Adana.

Sarıoğlu, T., Şengül H. (2024). Türkiye'de dış göç, tarımsal istihdam ve hanehalkı işgücü ilişkisi. *Tekirdağ Ziraat Fakültesi Dergisi, 21*(2), 348-361.

Shani, B. B., & Musa, A. (2021). The effect of mechanization on labour employment and cropland expansion in northern Nigeria. *AgroScience, 20*(3), 24-29.

Sevinç, G., Davran, M.K. ve Sevinç, M. R. (2018). Türkye'de Kırdan Kente Göç ve Göçün Aile Üzerindeki Etkleri. *İktsadi İdari ve Siyasal Araştırmalar Dergisi, 3*(6), 70-82.

TSI, (2025a). Agriculture- Agricultural land. https://data.tuik.gov.tr/Kategori/GetKategori?p=Tarim-111 (Accessed date: 15.06.2025)

TSI, (2025b). Agricultural Equipment and Machinery Statistics https://data.tuik.gov.tr/Kategori/GetKategori?p=Tarim-111 (Accessed date: 15.06.2025)

TSI, (2025c). Employment, Unemployment and Wages - Employed by Sector. https://data.tuik.gov.tr/Kategori/GetKategori?p=istihdam-issizlik-ve-ucret-108&dil=1 (Accessed date: 15.06.2025)

TZOB, 2023. https://www.tzob.org.tr/ulkemizde-tarim-nufusu-yaslandi/ (Accessed date: 23.06.2025)

Yalçın, G.E. ve Kara, F.Ö. (2016). Kırsal Göç ve Tarımsal Üretme Etkileri. *Harran Tarım ve Gıda Bilimleri Dergisi, 20*(2), 154-158.

Yıldız, D. Yıldız Çakır, N. Ilgaz, Y. Güneş, Ş. M. ve Yıldız, D. (2017). Türkiye Tarımı'nın Yapısal Geleceği. Türkiye'deki Tarımsal İşgücünün Demografik ve Yapısal Dönüşümü Projesi Ön Raporu, Su Politikaları Derneği Uygulama Araştırma Merkez, Ankara.

Yılmaz, E., Turğut, U., Tosun, D., Gümüş, S. (2020). İzmir İlindeki Çiftçilerin Kırsal Nüfusun Yaşlanma Eğilimi ve Tarımsal Faaliyetlerin Devamlılığına İlişkin Görüşleri. *Tarım Ekonomisi Dergisi, 26*(2), 109-119.

Yılmaz, E., & Günay, G. (2022). Göçmenler Ve Göçmen Ailelerin Karşılaştıkları Sorunlara İlişkin Bir Değerlendirme. *Journal of Karabuk University Faculty of Economics and Administrative Sciences, 2*(1), 1-19.

Yücel, M. H., & Çalışkan, Z. (2020). Tarımda verimlilik ve makineleşmenin tarımsal istihdam üzerindeki etkisi: Türkiye örneği. *Ekonomik Yaklaşım, 31*(117), 525-554. https://doi.org/ 10.5455/ey.17303

Appendix 1: Error Correction Model (ECM) Results

Variable	Coefficient	Standard Error	t-Statistic	p-value
C	0.003394	0.018753	0.181009	0.8594
D(LNYSEN)	0.472276	0.110928	4.257510	0.0011
D(LNYSEN(-1))	0.499075	0.194595	2.564683	0.0248
D(LNYSEN(-2))	0.406534	0.156162	2.603284	0.0231
D(LNTARALAN)	0.880649	0.836035	1.053363	0.3129
D(LNTARALAN(-1))	0.402393	0.633584	0.635106	0.5373
D(LNTARALAN(-2))	1.218437	0.421509	2.890651	0.0136
D(LNTRK)	1.670146	0.381320	4.379903	0.0009
D(LNTRK(-1))	1.619854	0.541836	2.989568	0.0113
ECT_1	-0.678336	0.109433	-6.198669	0.0000
R-squared	0.900342	Mean dependent var		-0.039020
Adjusted R-squared	0.825599	S.D. dependent var		0.045010
S.E. of regression	0.018797	Akaike info criterion		-4.807311
Sum squared resid	0.004240	Schwarz criterion		-4.311383
Log likelihood	62.88042	Hannan-Quinn criter.		-4.690485
F-statistic	12.04579	Durbin-Watson stat		2.656849
Prob(F-statistic)	0.000097			